Data Analyst Internship | Final Capstone

Predicting Diamond Prices with Machine Learning

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Problem and data

Problem

A retailer needs consistent, fast prices and a way to spot stones priced out of line. Goal: predict price from carat, cut, colour, clarity and dimensions.

Data

53,940 raw records cut to 53,790 clean ones: 146 duplicates removed, 20 zero dimensions imputed, 4 impossible records dropped.

Success criteria: R2 above 0.95 and average error well under 10%.

Approach

1

Features: ordinal cut, colour and clarity, plus volume = x * y * z

2

Target: log(price), because price is right-skewed; results converted back to dollars

3

Three models on one 80/20 split: linear regression, random forest, gradient boosting

Results

0.982
R2 on unseen test data

57
Average error (MAE)

6.2%
Average error in percent

Gradient boosting and random forest tie at R2 0.982; the linear baseline reaches 0.944 with an 11.4% error. Five-fold cross-validation gives R2 0.993 on log price.

Where the model errs

81%

of predictions within 10% of the actual price

97%

within 20%

9.4%

average error for stones above 2 carat, against 5.6% under 0.5 carat

Dollar errors grow with size, from about 8 to ,316. The worst single misses are over 60%, so unusual stones need expert review.

Insights

- Size drives price: carat and dimensions dominate feature importance.
- Clarity is the next most important feature, then colour.
- Cut adds almost nothing once size is known.
- Depth and table have very little effect.

Recommendations and next steps

- Use the model as a pricing guide and to flag stones far from the predicted price.
- Send large or unusual stones to expert review.
- Limits: one snapshot, no sales or margin data, no certification details.
- Next: add sales and date data, tune the model, and monitor errors over time.